IS-EGR
Endocrine-1 Module

ENDOCRINE PHYSIOLOGY

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Endocrine Physiology Study Resources (Canvas)

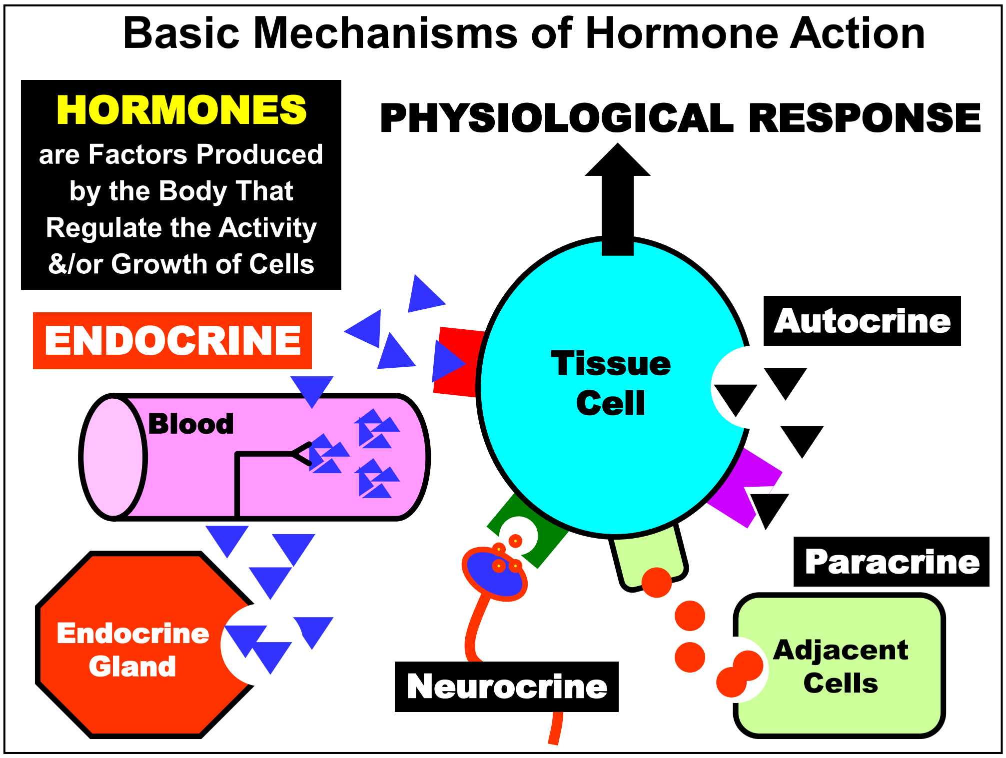
1. **Endo Physiology Study Guide Notes**
2. **Endo Physiology Exam Objectives**
3. **Self-Study Narrated PPt Presentations**
4. **Optional Live Zoom Q & A Sessions**
5. **Endo Physio Practice Exam Questions**
6. **Endo Basic Science Quiz (Canvas Assignment)**
7. **Virtual Office Hours (TBA)**

Narrated PowerPoint Presentations

1. **Overview & Principles of Endocrine System**
2. **Hypothalamic Neuroendocrine Hormones**
3. **Anterior Pituitary Hormones**
4. **Posterior Pituitary Hormones**
5. **Thyroid Hormones Basic Properties**
6. **Thyroid Hormone Disorders**
7. **Parathyroid Hormone**
8. **Vitamin D and Calcitonin**
9. **Adrenal Cortex Hormones**
10. **Steroid Hormone Biosynthesis & CAH Disorders**
11. **Adrenal Medulla Hormones**

Introduction to Endocrine Physiology:

OVERVIEW & PRINCIPLES

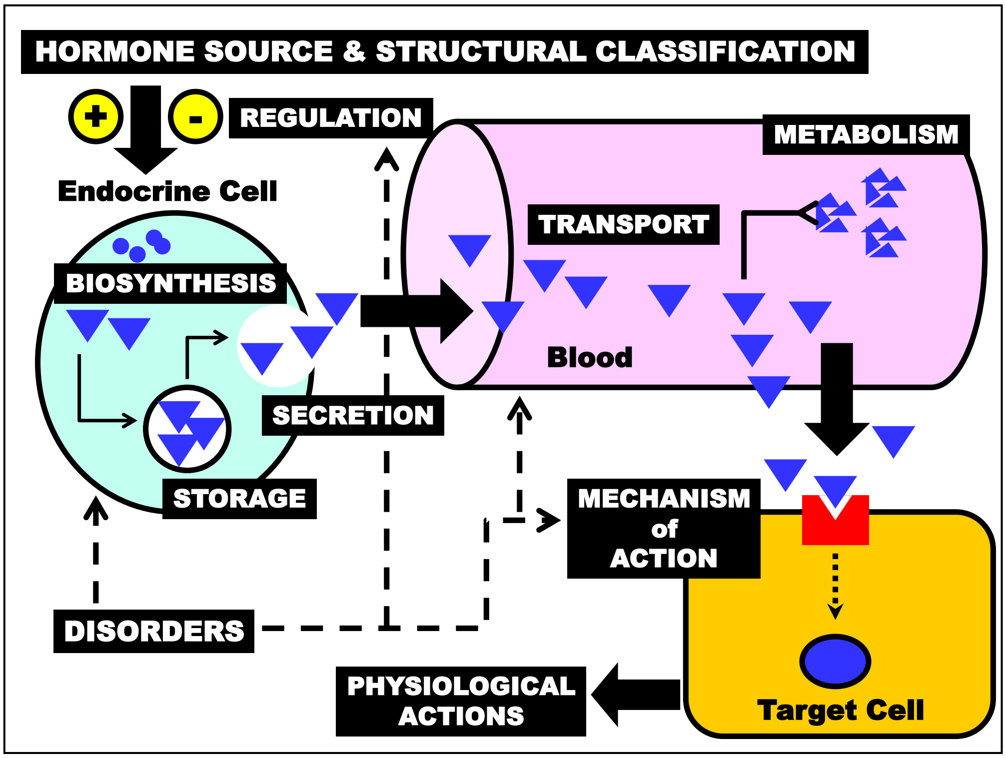


Hormones are factors produced by the body to regulate the activity &/or growth of cells (detailed later).

Various mechanisms are used by Hormones to elicit a Physiological Response from Tissue Cells:

For example, Autocrine, Paracrine (*left*) and Juxtacrine (*not shown*) mechanisms are generally used for "Local" control of a few or single Tissue Cells. Hormones and other hormone-like regulatory factors that act via these mechanisms are typically released from cells located near the responding Tissue Cells. Neurocrine factors are released from nerve endings that synapse with Target Cells. Neural inputs may originate from a distance (spinal cord) or locally (brain and enteric neurons) to affect a few or many cells.

The primary focus of this section will be on the **Endocrine** mechanism (left) of regulation in which hormones are released into the **Blood** from **Endocrine Glands** or cells located a far distance from Target Cells. Endocrine hormones often regulate multiple functions related to many cells of many different tissues.

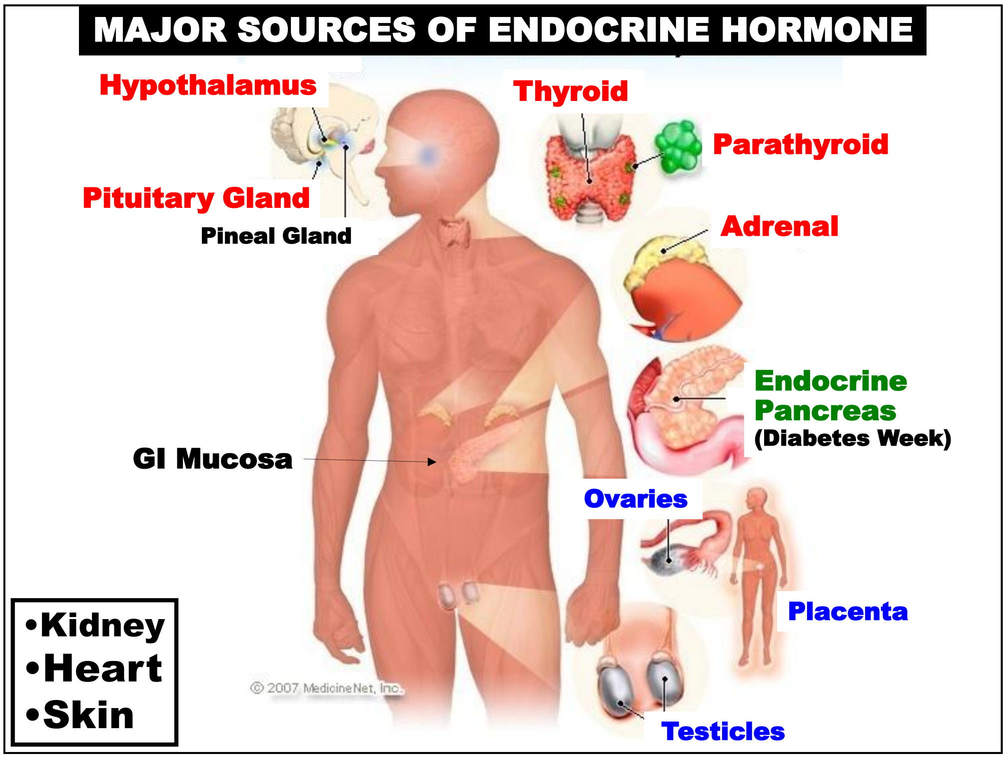


The basic properties of endocrine hormones that relate to their physiological functions and actions are illustrated above.

Each property will be discussed in relation to the three structural classes before introducing specific hormones.

- **Major Sources of Endocrine Hormones**
- **Specific Endocrine Hormones & Source**
- **Hormone Structural Classification**
- **Biosynthesis, Secretion, Storage**
- **Transport in Blood**
- **Metabolism**
- **Mechanism of Action**
- **Regulation of Secretion**
- **Physiological Actions**
- **Clinical Disorders: Pathophysiology**

These are the specific properties of most of the endocrine hormones listed in Tables 1,2, 3 that will be presented and that you will be expected to distinguish for assessment purposes



The basic scheme outlined in the two previous figures will be applied to our discussion on the Endocrine Organs indicated above.

Our focus will be on the following glands:

Hypothalamus
Pituitary Glands
Thyroid Glands
Parathyroid Glands
Adrenal Glands

Endocrine Pancreas (presented during Diabetes Week)

GI Hormones were discussed in earlier units and hormones of the Ovaries & Testes will be discussed later during Reproduction Module.

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- **Specific Endocrine Hormones & Source**
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- **Biosynthesis, Secretion, Storage**
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- **Mechanism of Action**
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- **Physiological Actions**
- **Clinical Disorders: Pathophysiology**

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Table 1 Hypothalamic & Pituitary Hormones

Hormone	
Corticotropin Releasing H. (**CRH**)*	Hypothalamus
Thyrotropin Releasing H. (**TRH**)	
Gonadotropin Releasing H. (**GnRH**)	
Growth Hormone Releasing H. (**GHRH**)	*Know Hormone Abbreviations
Somatostatin (**GHRIH**)	
Prolactin Releasing Factors (**PRF**)	
Prolactin Release Inhibiting H. (**PRIH**)	
Adrenocorticotropic H. (**ACTH**)	Anterior Pituitary
Thyroid Stimulating H. (**TSH**)	
Leutinizing Hormone (**LH**)	
Follicle Stimulating H. (**FSH**)	
Growth Hormone (**GH**)	
Prolactin (**PRL**)	
Vasopressin (**ADH**)	"Posterior Pituitary" (Hypothalamus)
Oxytocin (**OT**)	

The next three figures summarize the Major Endocrine Hormones according to their Sites of Secretion.

You should **know the structural classification AND abbreviation for most of these hormones** that will be discussed in this section.

The rationale is that most hormones are referred to by their abbreviation on clinical laboratory reports.

Table 2 Peripheral Endocrine Hormones

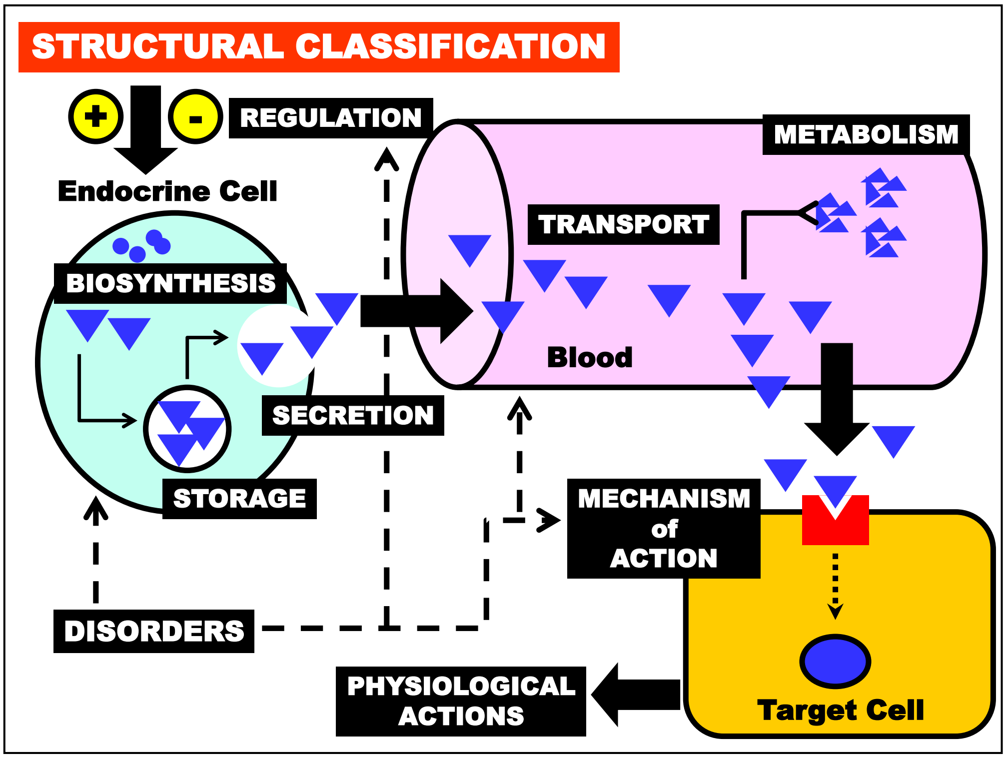
Hormone	Site of Secretion
Thyroxine (T4)	Thyroid Gland
Triiodothyronine (T3)	Thyroid Gland
Calcitonin	Thyroid Gland
Cortisol	Adrenal Cortex
Aldosterone	Adrenal Cortex
Androgens & Estrogens	Adrenal Cortex & Gonads
Epinephrine & Norepinephrine	Adrenal Medulla
Insulin	Endocrine Pancreas
Glucagon	Endocrine Pancreas
Somatostatin	Pancreas, Hypothal., GI
Parathyroid Hormone	Parathyroid Gland
Renin	Renal Tubules
Angiotensin	Liver/Lung
Vitamin D (Calcitriol)	Skin/Liver/Kidneys

More Hormones.

Table 3 Other Endocrine Hormones

Hormone	Site of Secretion
Gastrin, Secretin, CCK, GIP, Motilin	GI Tract
Estradiol Progesterone Inhibin	Ovaries
Testosterone Inhibin	Testes
Estrogens & Progesterone Human Chorionic Gonadotropin Human Chorionic Somatomammotropin	Placenta
Erythropoietin	Kidney
Insulin-Like Growth Factor-1	Liver
Melatonin	Pineal
Melanocyte Stimulating Hormone	Intermed Pit.

Even more Hormones



The basic properties of endocrine hormones that relate to their physiological functions and actions are illustrated above.

Each property will be discussed in relation to the three structural classes before introducing specific hormones.

Structural Classification of Hormones

Hormones Within a Structural Class Share Certain Basic Physiological Properties

- **PEPTIDES & PROTEINS**
  - **MOST HORMONES**
  - Insulin, Glucagon
  - Gastrin, Secretin, CCK
  - ADH, Renin
- **STEROIDS (Cholesterol)**
  - Glucocorticoids (Cortisol)
  - Mineralcorticoids (Aldosterone)
  - Reproductive Steroids (Testosterone)
- **AMINES & AMINO ACIDS** (i.e. **Tyrosine**)
  - Thyroid Hormones (Thyroxine)
  - Catecholamines (Epinephrine)

FATTY ACIDS
- Eicosanoids
- Retinoids

(not discussed)

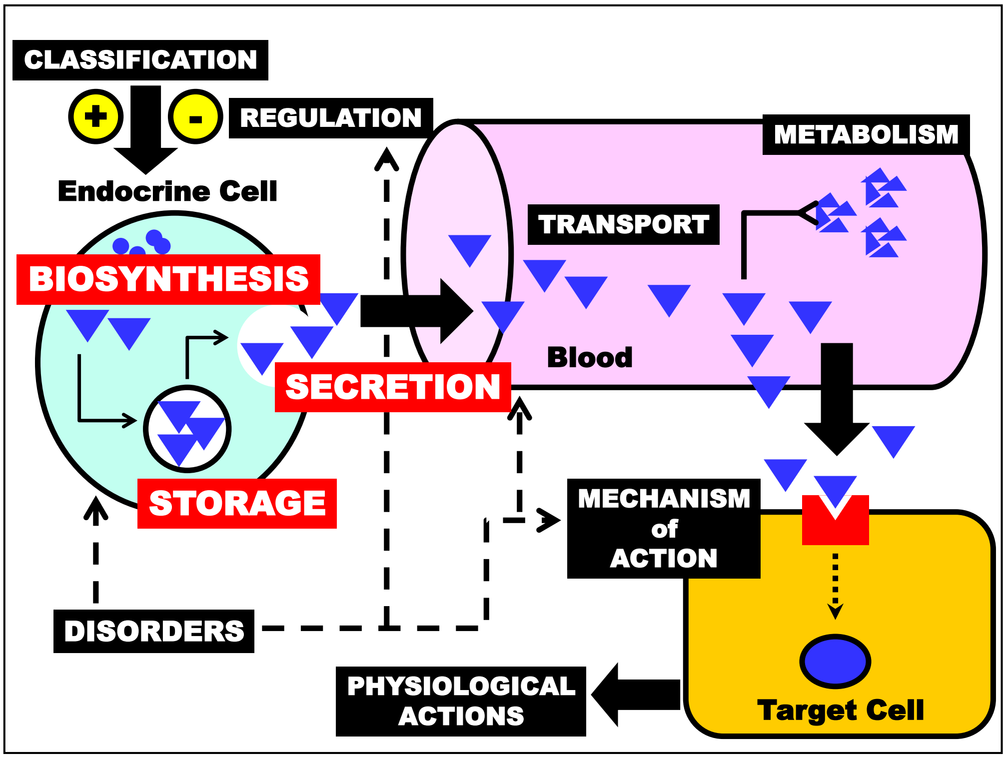
Endocrine hormones are classified into 3 principle categories based on their **Biochemical Derivation**.

By far, Most Hormones are either small **Peptides** or larger **Proteins**. The other major hormone class is the **Steroids**, which are structurally derived from **Cholesterol**. Finally, a relatively small group of hormones are derived from single or coupled amino acids (**Amino Acid Derivatives**). Some general examples of each are listed above.

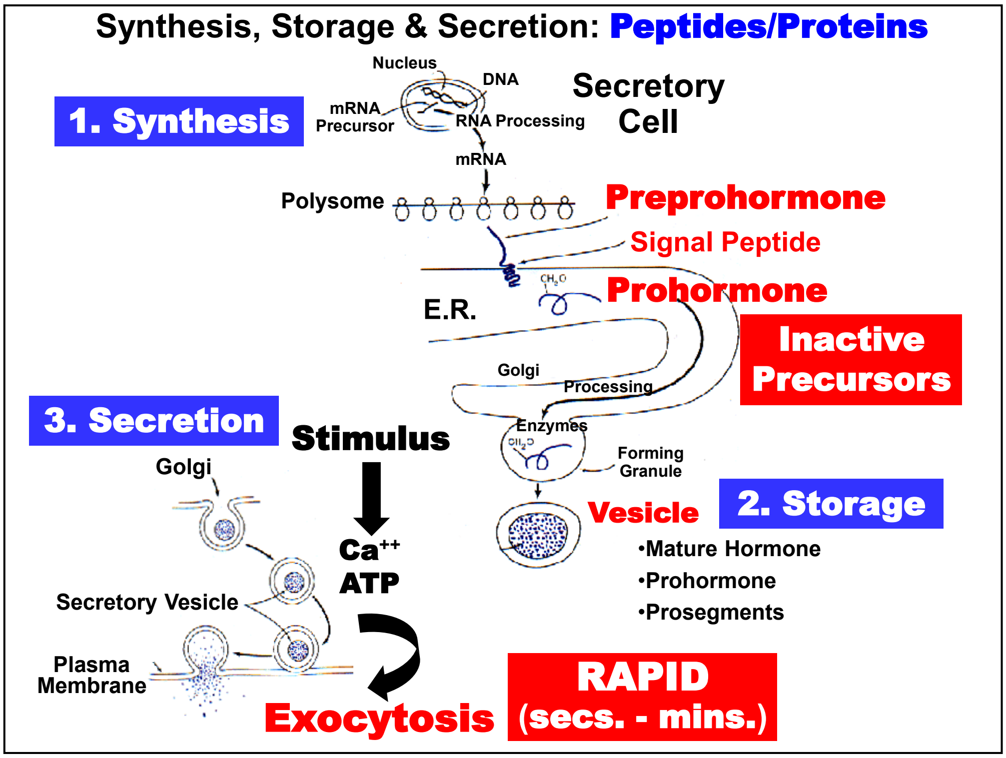
A fourth class of regulatory hormones derived from **Fatty Acids** *(bottom right)* will not be discussed here as they mainly serve in mediating and/or facilitating hormone action. Nevertheless, these are important molecules pharmacologically as they are often the target of many drug actions.

Knowledge of hormone chemical classification is important towards organizing the many basic properties of each specific hormone.

That is, **hormones within a structural class share certain basic physiological properties**. Knowledge of these general class principles obviates the need to memorize an individual set of properties for each hormone.



Hormone **<u>Synthesis, Storage & Secretion</u>** in blood is discussed next.

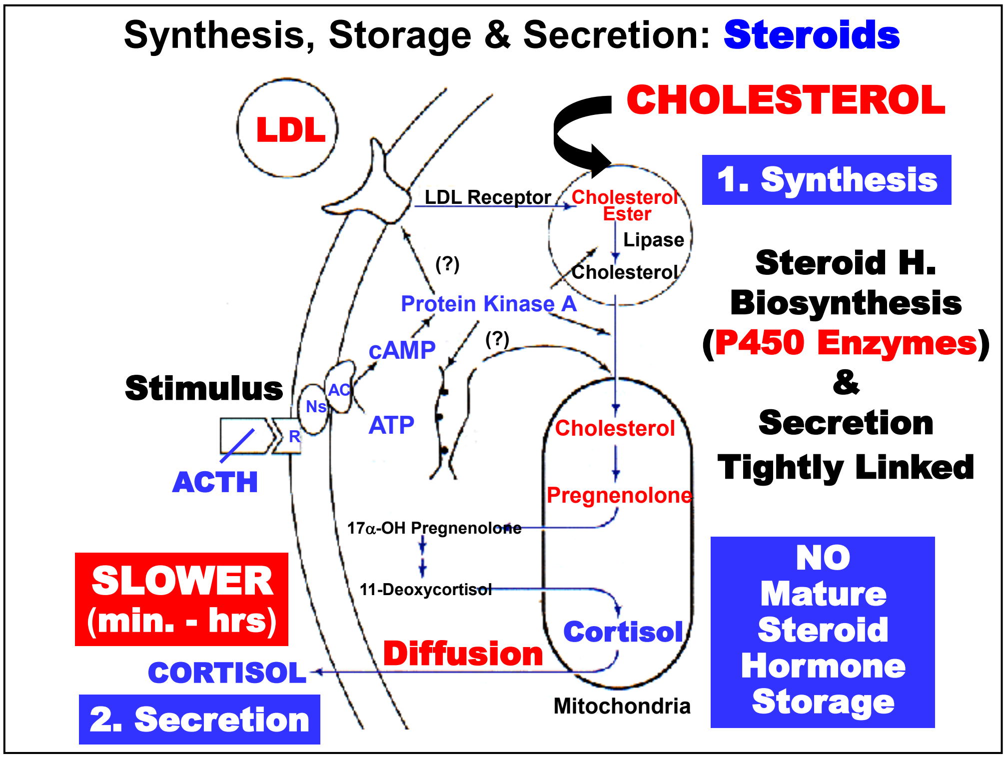


1. **Synthesis**. **Protein Hormones** are synthesized within secretory cells (*top*) in a manner typical of most cellular proteins. However, most are synthesized initially as larger Inactive Precursors termed **Preprohormones**. The lipophilic "Pre" segment (or signal peptide) facilitates extrusion of polypeptides into the ER where it is cleaved by a peptidase.

2. **Storage**. Resulting **Prohormones** are transported to the Golgi where they are packaged, cleaved to the active mature hormone form and stored in Secretory Vesicles until released in response to a stimulatory signal.

3. **Secretion**. Protein hormones are typically released from secretory cells by a Ca++ & ATP-dependent process termed **Exocytosis**. **Mature Hormone, Prosegments & Prohormone may all be secreted.** Relative levels are sometimes used clinically to assess synthesis and secretion rates when a deficiency or hypersecretion disorder develops.

An important concept is that protein hormone secretion is normally quite **RAPID**, requiring only **seconds-minutes** to enter the blood in response to a secretion stimulus.



1. **Biosynthesis**. **Steroid Hormones** are synthesized from the common steroid precursor, **Cholesterol**, via a series of enzymatic modifications involving a family of **P450 Enzymes**. (*top*) Plasma lipoproteins (LDL) appear to be the major source of cholesterol although Cholesterol Ester (intracellular stored cholesterol) is also utilized. A key enzymatic step in steroidogenesisis is the conversion of Cholesterol to Pregnenolone within Mitochondria. Subsequent enzymatic modifications result in a mature steroid secretory product (e.g. Cortisol).

An important concept is that in contrast to peptide hormone storage in secretory granules, there is **NO Mature Steroid Hormone Storage** intracellularly. Thus, for steroid hormones, the processes of Synthesis & Secretion are Tightly Linked. That is, when a secretory Stimulus (e.g. ACTH *middle left*) is received by the cell, *de novo* hormone Synthesis is required *prior* to hormone Secretion. Consequently, steroid hormone secretion is relatively **SLOW** (**Minutes-Hours**) compared to protein hormones.

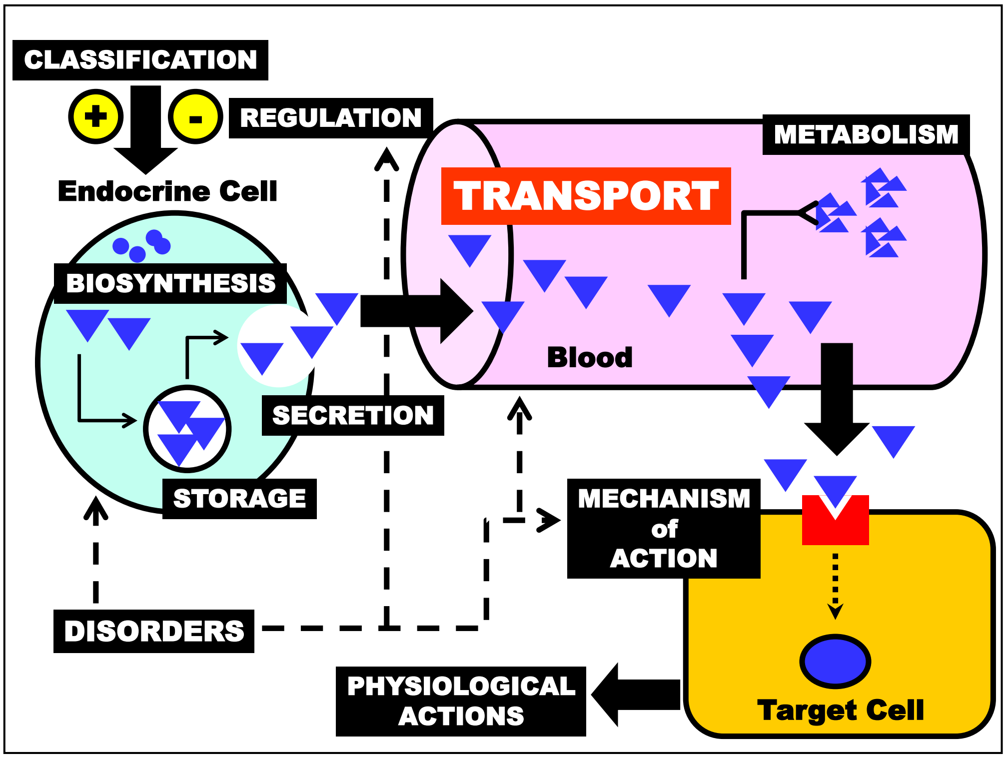
2. **Secretion**. Steroid & protein hormones also differ in their mechanism of secretion into blood. Due to their High Lipid Solubility, steroid hormones are secreted by **DIFFUSION** across the secretory cell membrane into the blood immediately following synthesis.

Synthesis, Storage & Secretion:
Amino-Acid Derived

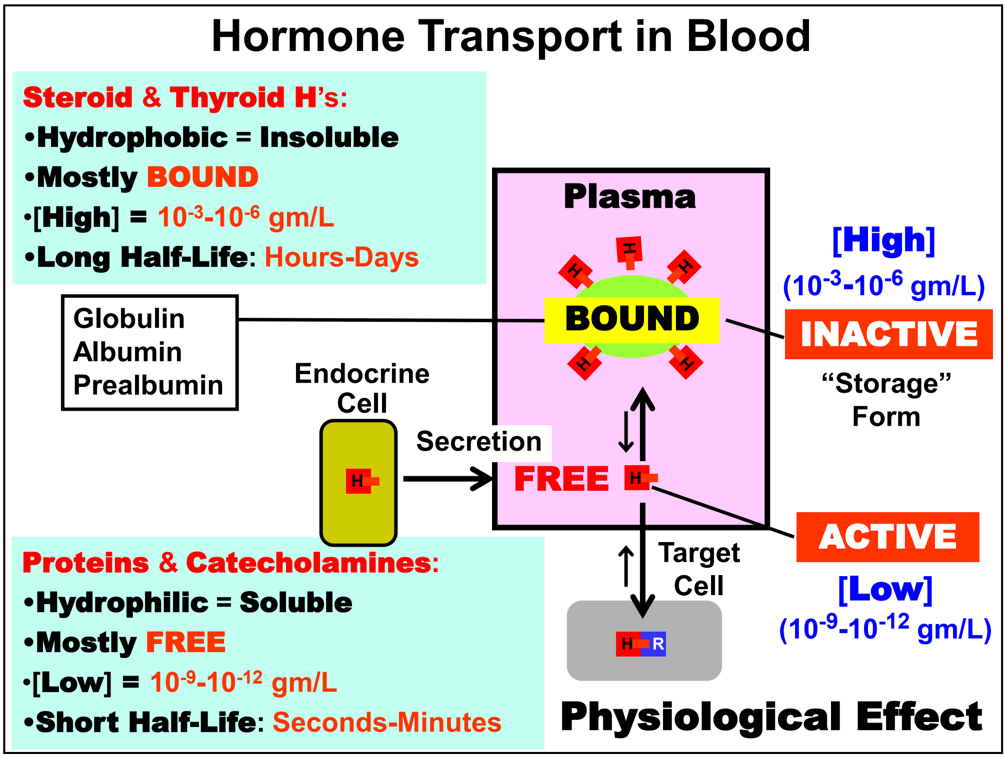
Unique to Each Hormone in Class:

- **Thyroid Hormones**
- **Adrenal Catecholamines**
- **Prolactin Inhibiting Hormone (Dopamine)**

Bioynthesis, Storage and Secretion of the principle **Amino-Acid Derived** hormones are relatively unique to each hormone and will be discussed separately in relation to the specific hormone.



Hormone **Transport** in the blood is discussed next.



Hormones are carried in the blood in two forms: "**Free**" and "**Bound**"

Most **Protein Hormones & Catecholamines** are **Hydrophilic** and thus, relatively **Soluble** in plasma. These types of hormones predominantly exist in what is termed the **FREE** or soluble form. This basically means that they are Dissolved in plasma. There are a few exceptions within these groups of hormones, but this is the general rule.

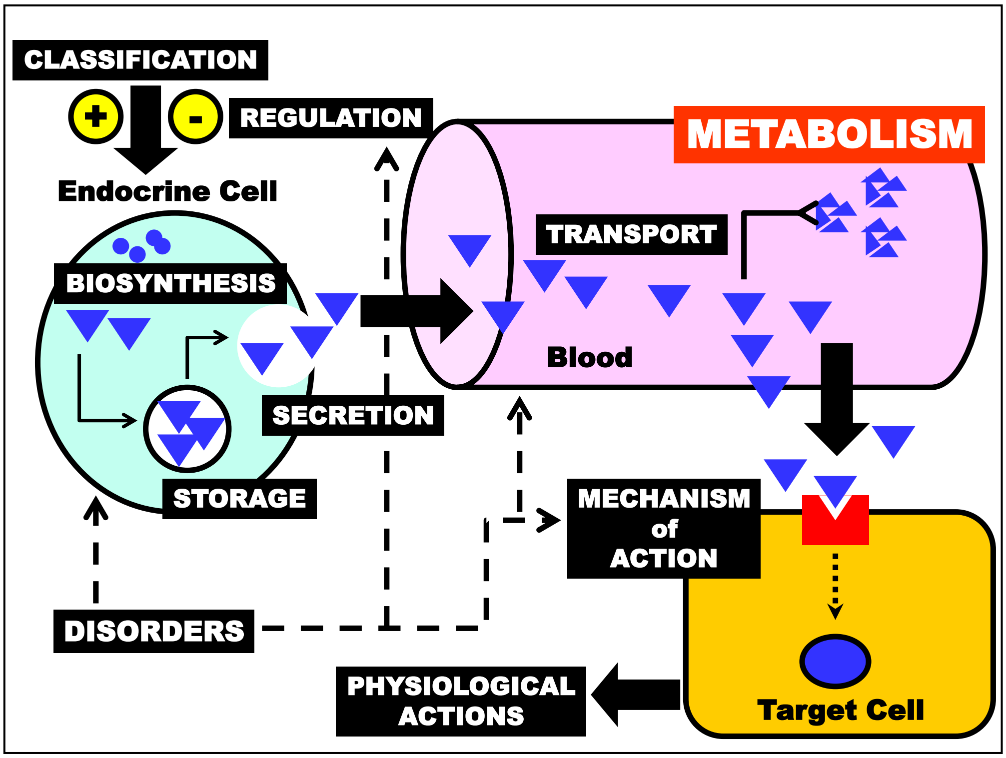
An important concept is that the FREE form of a hormone is considered the **Physiologically ACTIVE** form of the hormone because only dissolved hormone is capable of binding to Target Cell Receptors to stimulate a Physiological Response.

Because almost all Protein & Catecholamine hormone circulating in the blood is in the FREE active form, their normal steady-state concentration in blood can usually be relatively **LOW** (nanogram-picogram/Liter range) in order to maintain homeostasis of the parameter(s) they regulate.

By contrast, most **Steroid & Thyroid Hormones** are **Hydrophobic** and thus, relatively **Insoluble** in plasma. These types of hormones must circulate predominantly **BOUND** to plasma proteins (Globulins, Albumin, Prealbumin) that act to solubilize the hormones in plasma.

However, a small proportion of the **FREE** form of the these hormones also exists in equilibrium with the much larger Bound fraction. This free fraction is important in that it serves as the physiologically "Active" Pool of the hormone that is capable of interacting with Target Cell Receptors. The bound fraction is also important because it serves as a circulating "Storage Form" of the hormone in the event that more hormone is needed to maintain homeostasis of a regulated parameter.

Because of their limited solubility, Steroid & Thyroid Hormones must typically circulate at **HIGH** concentrations (milligram-microgram/Liter range) in order to maintain an adequate level of the FREE active form of the hormone in the blood.



Hormone **Metabolism** is discussed next.

Hormone Inactivation & Excretion

HOW?:

- **Metabolized** (Degraded)
  - **Proteases**: Most Protein H.
  - **Deiodinases**: Thyroid H.
- **Modified:** (Structure Preserved)
  - H-CH_3; H-OH; Red./Ox.
  - Glucuronate; H-SO3
  - Purpose = Solubilization
    - Excretion: Urine
  - Most Steroid H. & Catecholamines

WHERE?

LIVER
&/or
KIDNEY
&/or
TARGET CELLS

Disorders Related to These Organs May ↑↓ [HORMONE]

(e.g. Cirrhosis)

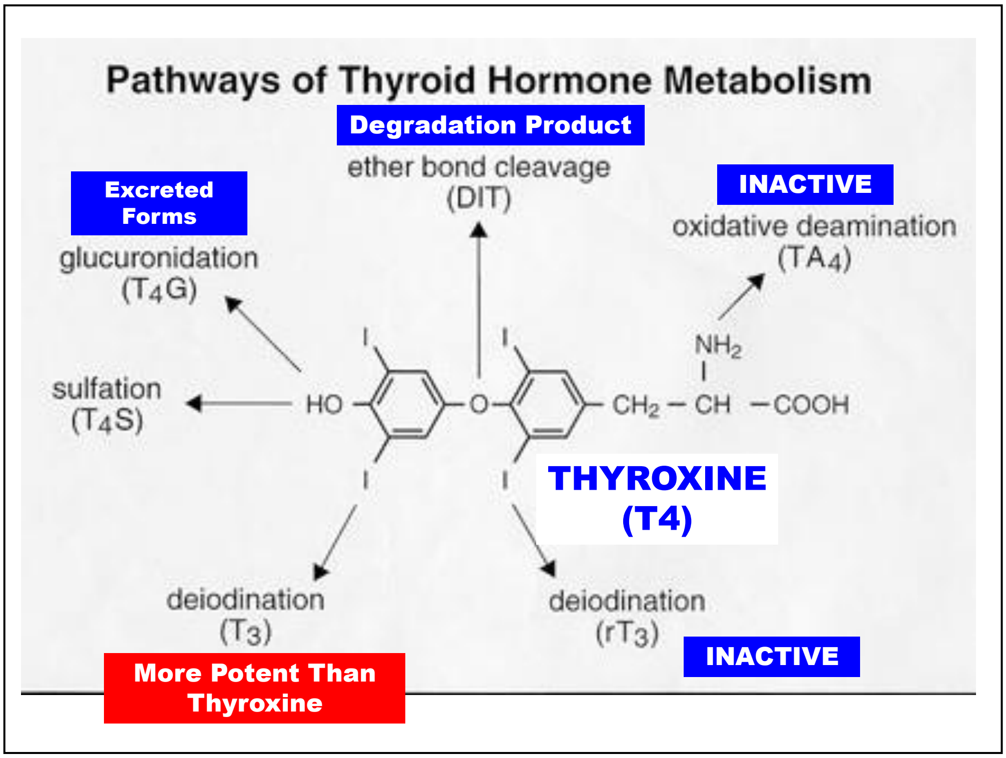
Inactivation of Hormones must occur following their action on target cells in order to cease cell stimulation and to re-establishe its normal plasma level.

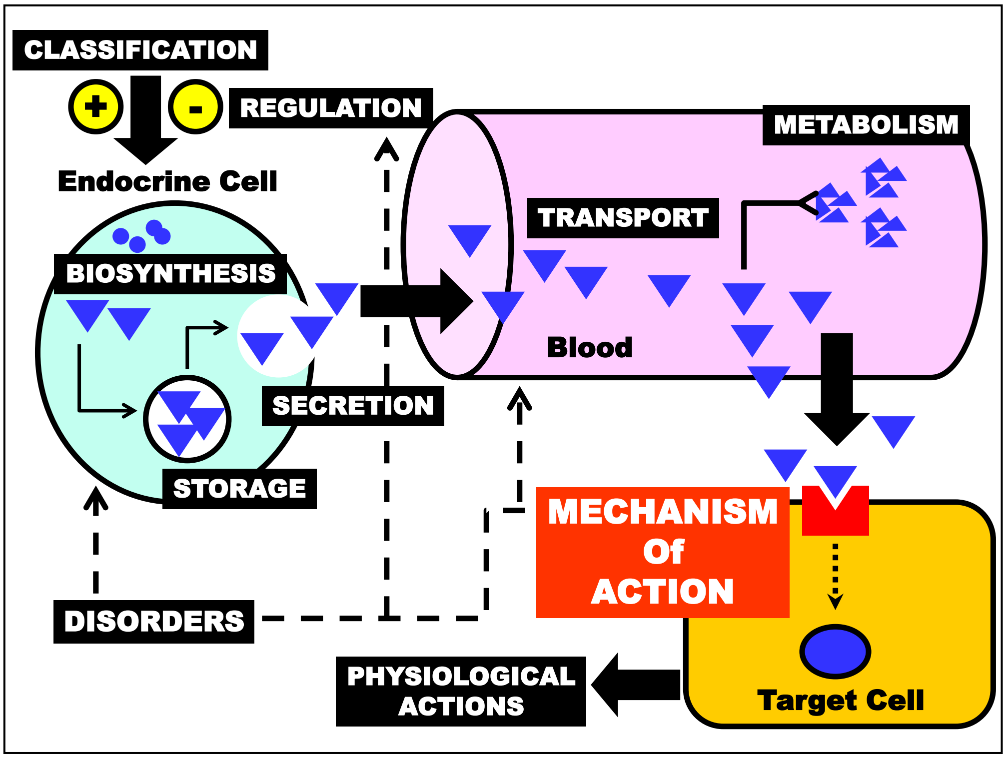
WHERE? Major sites of metabolic inactivation of most hormones are the **Liver & Kidneys** (also Target Tissues). Thus, Liver & Kidney Disorders Often Disrupt Normal Plasma Hormone Levels.

HOW? Inactivation occurs through two basic mechanisms:

Many hormones are **Metabolized** (Degraded), leaving no recognizable by-product. Hydrolysis by plasma or cellular **Proteases** is the most common mechanism and is the primary mechanism used to inactivate **Most Protein Hormones**. **Deiodinases** are specialized enzymes used to degrade the iodine-containing **Thyroid Hormones**.

Other hormones are inactivated through **Biochemical Modification** [Sulfated (SO3-), Hydroxylated (OH-)], Conjugation (Glucuronic Acid) in which the hormone Structure is Preserved. **Most Steroid Hormones** are inactivated in this manner because of the inability to degrade cholesterol core. Amino acid-derived **Catecholamines** are another example. Some modifications also solubilize the inactive hormone for **Excretion In Urine**. By-products are often used clinically to estimate hormone level & production.


Pathways of Thyroid Hormone Metabolism
Degradation Product
ether bond cleavage
(DIT)
Excreted Forms
glucuronidation
(T_4G)
INACTIVE
oxidative deamination
(TA_4)
sulfation
(T_4S)
HO
O
I
NH_2
CH_2 — CH — COOH
THYROXINE (T4)
deiodination
(T_3)
deiodination
(rT_3)
INACTIVE
More Potent Than Thyroxine



Hormone **Receptors** (Mechanism of Action) on Target Cells are discussed next.

Endocrine Hormone Receptor Types

MEMBRANE (Surface) Receptors:
- **G-Protein Coupled**
- **Tyrosine Kinase**

Peptides Protein H. & Catecholamines

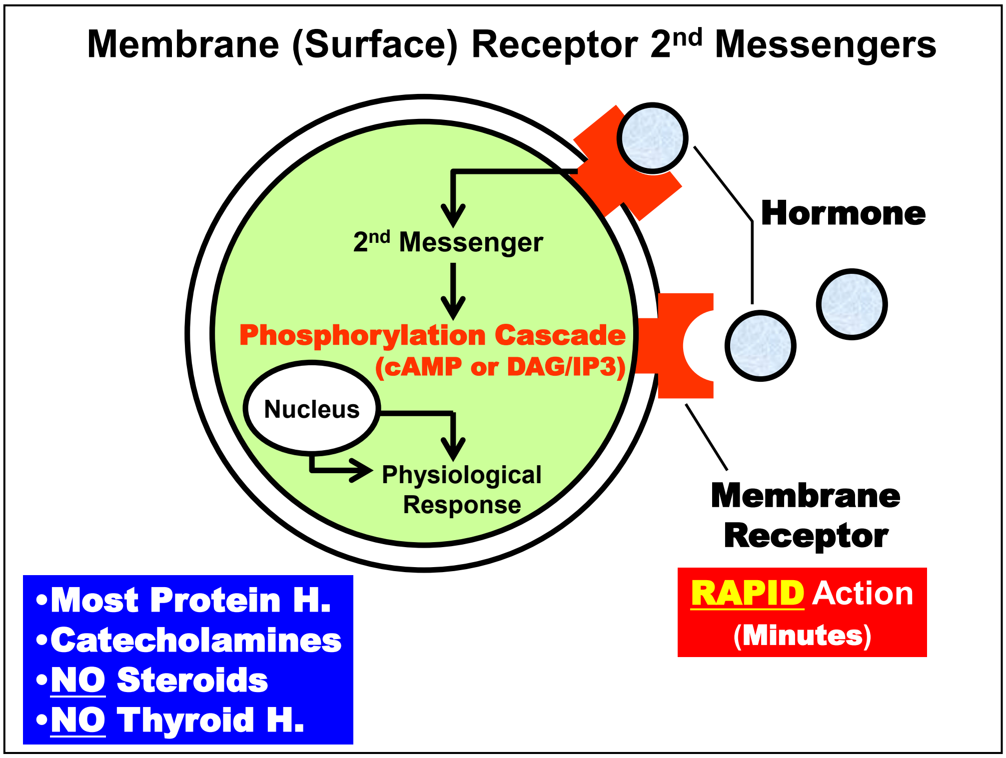
INTRACELLULAR Receptors:
- **Cytoplasm**
- **Nucleus**

Steroid & Thyroid Hormones

Hormone action occurs through their binding of **Target Cell RECEPTORS**. Receptors function to Bind Hormones with high specificity to target cells, which initiates a cascade of biochemical events via various 2nd Messenger Signaling Pathways that elicit specific biological responses from the cell. Two basic classes of hormone receptors are recognized.

Membrane or Surface Receptors are located on the **Cell Membrane Surface**. Two major families of membrane receptors are known: G-Protein Coupled Receptors, Tyrosine Kinase Coupled Receptors. **Protein Hormones** & **Catecholamines** act, in all known cases, through one or more of these types of receptors. Steroid & Thyroid Hormones do **NOT** act significantly via membrane receptors.

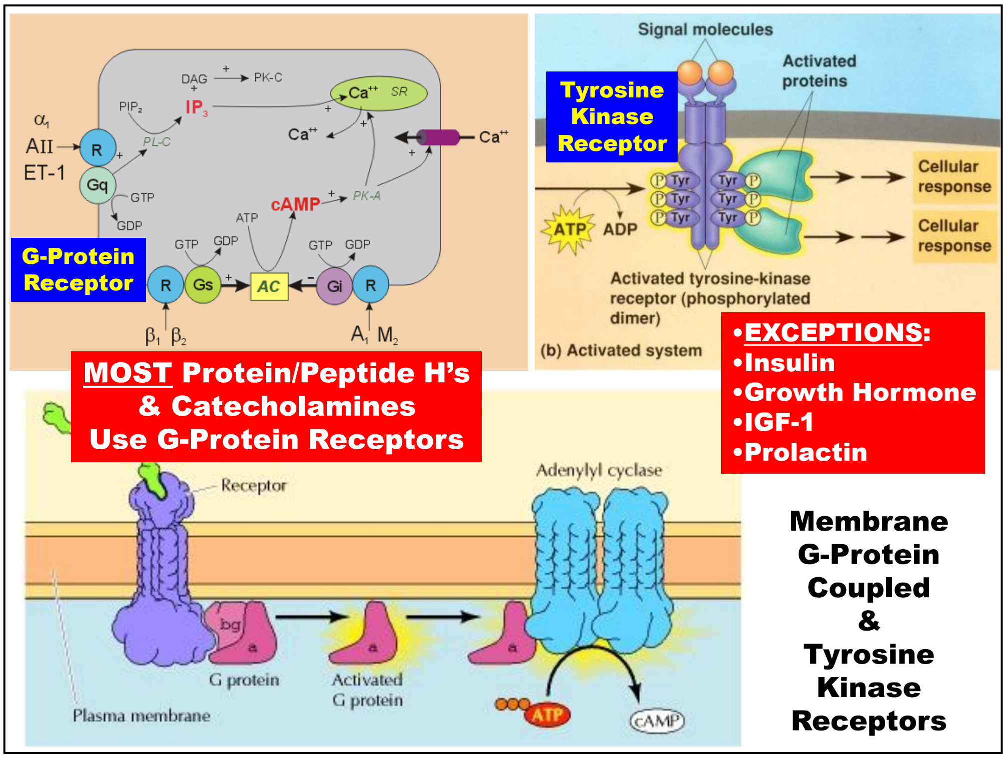
Intracellular Receptors are located **Intracellularly** in the target cell Cytoplasm or Nucleus. **Steroid** & **Thyroid Hormones** act predominantly through this type of receptor.



Binding of Hormone to **Membrane Receptors** typically elicits a Physiological Response from Target Cells by stimulating a 2nd Messenger Pathway the results in a **Phosphorylation Cascade** (e.g. **via cAMP or DAG/IP3**). Phosphorylation functions to activate or inactivate a series of proteins related to the function of the cell and action of the hormone. Less often, phosphorylation stimulates the gene transcription and synthesis of proteins (Nucleus) related to the function of the cell and action of the hormone.

Because Phosphorylation events are typically transduced very quickly within cells, **Protein Hormones & Catecholamines** usually exert a **RAPID** action on Target Tissues. That is, the Physiological Response from target cells usually occurs within **Minutes** after Hormone binding to receptors.

This is in contrast with the action of Steroid & Thyroid Hormones described next.



Summary of cell biology of membrane receptor mechanisms:

There are two subtypes of membrane receptors: **G-Protein Coupled** (top left) and **Tyrosine Kinase Receptors** (top right). Most Protein/Peptide hormones and the Catecholamines act via G-Protein receptors.

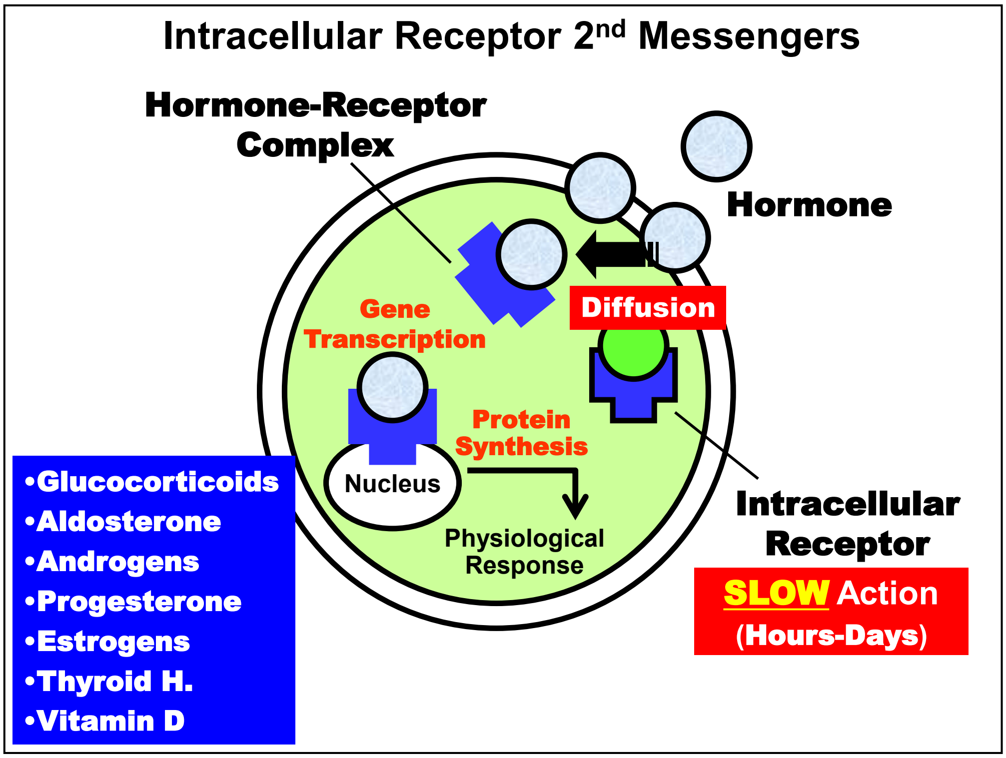
However, notable exceptions are **Insulin**, **Growth Hormone, IGF-1 (Insulin-Like Growth Factor)** and **Prolactin**, which regulate target cells via a **Tyrosine Kinase Receptor** mechanism.

NOTE: Be able to distinguish these four hormones from other hormones acting via G-protein receptor types.

(top left) G-Protein coupled receptors are further distinguished in terms of their second messenger mechanisms: **Diacylglycerol/Inositol Triphosphate (DAG/IP3)** and **Adenylate Cyclase/cAMP** pathways.

NOTE: You will not be asked to distinguish hormone based on their second-messenger mechanism although you should be aware of the different pathways and that they generally initiate a **Phosphorylation Cascade** that mediates its cellular action.

(bottom left) The cAMP pathway is detailed.



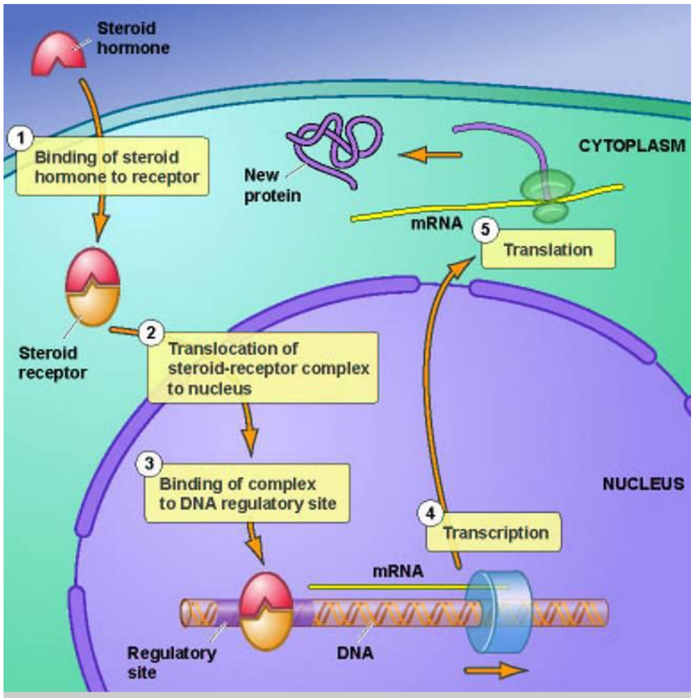
Binding of hormones to **Intracellular Receptors** requires that the hormone first enter the cell across the cell membrane. Because of their Hydrophobic nature, Steroid & Thyroid Hormones predominantly move into the cell by **Diffusion** and subsequently bind to a pool of intracellular receptors maintained in the cytoplasm or nucleus.

Hormones acting through intracellular receptors typically elicit a Physiological Response from Target Cells by stimulating **Gene Transcription** and **Protein Synthesis** of gene products related to the function of the cell and action of the hormone. Contrast this with Protein Hormones & Catecholamines.

Because Gene Transcription & Translation are typically slow processes, **Steroid & Thyroid Hormones** usually exert a **SLOW** action on Target Tissues relative to Proteins & Catecholamine. That is, the Physiological Response from target cells usually requires **Hours** and sometimes **Days** to develop after Hormone binding to receptors.

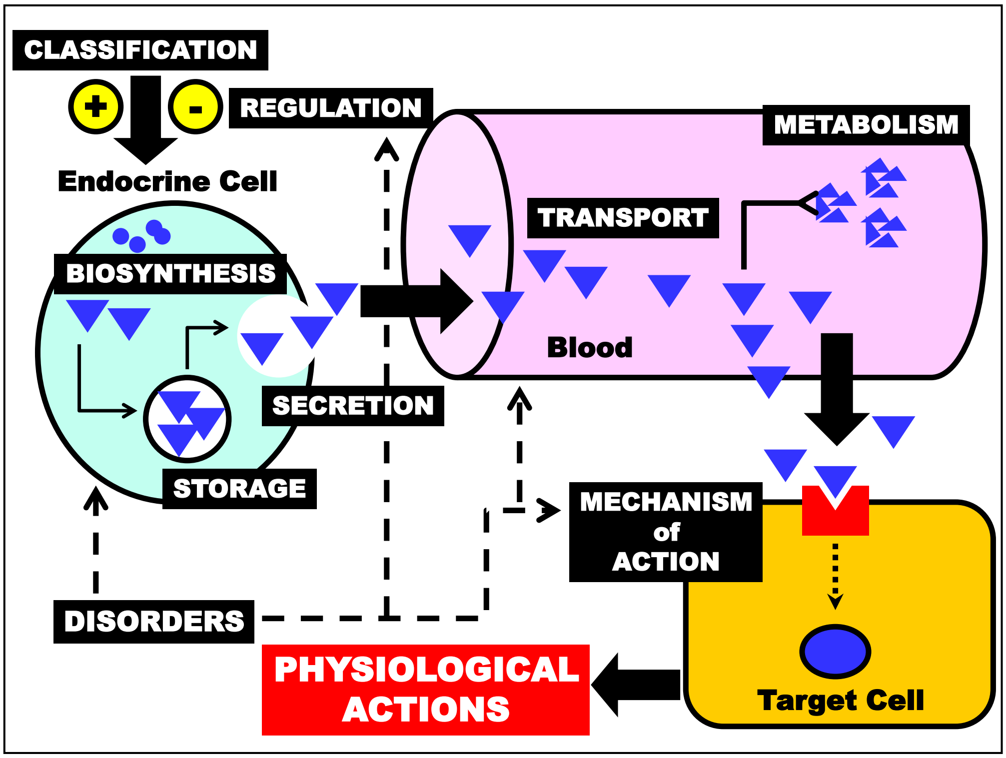
This property is important clinically because this delay must be accounted for in patients placed on Steroids pharmacologically. For example, patients who have elective surgery are often given Steroids to reduce post-surgical inflammation a few days prior to surgery.

Intracellular Steroid Hormone Receptor Mechanism

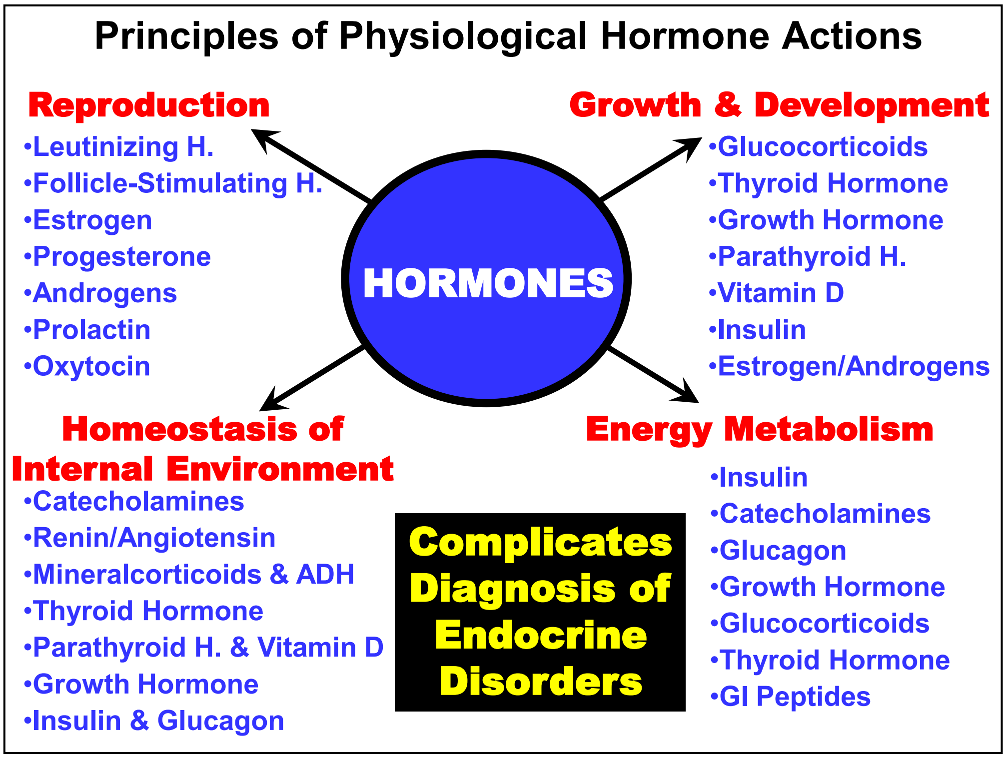


Source: "Module 6," < http://www.thepepproject.net/ >

Summary of cell biology of intracelluar receptor mechanism.



Hormone **Physiological Actions** are discussed next in general terms.



The fundamental physiological processes (e.g . **Energy Metabolism**) regulated by endocrine hormones are outlined above in relation to some of the hormones known to be involved in their regulation.

Unlike other hormone properties discussed earlier, the **physiological action(s) of each hormone are unique to each hormone**.

Note that the integrated actions of <u>multiple hormones are typically involved in regulating a single process</u> (e.g. Reproduction). Hormones involved may exert similar or overlapping effects on a single process (e.g. Catecholamines & Glucagon actions on Glycogenolysis). Conversely, other hormones exert antagonistic actions on a the same physiological process (e.g. Insulin & Glucagon actions on Glycogenesis).

Also note that there are a single hormone may play a role in several different homeostatic processes. For example, the thyroid hormones play a regulatory role in energy metabolism, growth and development, and homeostasis.

Thus, <u>**understanding the pathophysiology and clinical manifestations of endocrine disorders is complicated**</u> because of the integrated nature of hormone action on physiological processes (*discussed next*).

Hormones Maintain PHYSIOLOGICAL HOMEOSTASIS (FEEDBACK CONTROL)

- Catecholamines
- Renin/Angiotensin
- Mineralcorticoids & ADH
- Thyroid Hormone
- Parathyroid H. & Vitamin D
- Growth Hormone
- Insulin & Glucagon

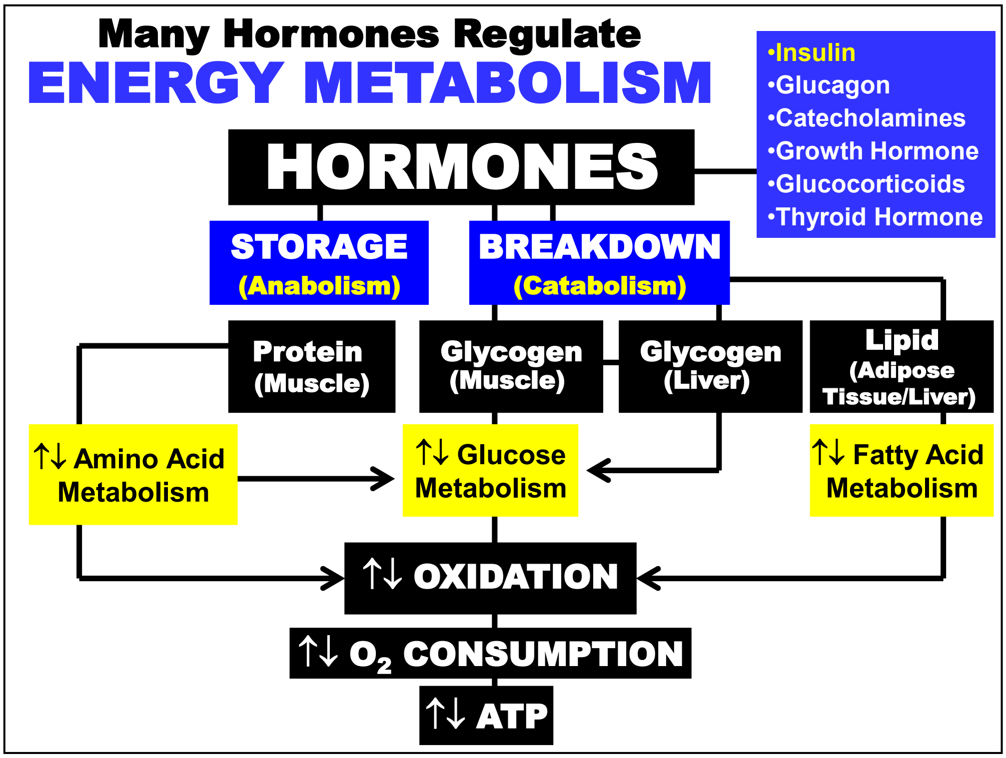
•Blood Pressure	120 mmHg/80 mmHg
•Blood pH	7.35-7.45
•Plasma Osmolality	280-296 mOs
•Plasma Electrolytes:	
•[Na^+]	135-145 mM
•[K^+]	3.5-5 mM
•[Ca^{++}]	2.1-2.6 mM
•Energy Metabolites:	
•[Glucose]	60-100 mg/dL
•[Triglycerides]	40-150 mg/dL
•etc.	

Hormones are involved in maintaining **Physiological Homeostasis**.

All physiological parameters are maintained within a normal physiological range (examples are shown above). Hormones function to maintain these values whenever they deviate significantly from these ranges due to various physiological or pathological conditions.

Regulation usually involves some type of a “**FEEDBACK**” control mechanism and will be detailed later.

The central and autonomic nervous systems act in concert with endocrine hormones to maintain physiological homeostasis.



Hormones play a major role in maintaining **Energy Metabolism** and production.

Many hormones act on the various energy substrate stores (Protein, Lipid, (Carbohydrates) Glycogen) in the body to either store (Storage) excess fuel substrates or to mobilize and metabolize (Breakdown) them via Oxidation for energy generation (O2 Consumption and ATP).

Energy substrate <u>storage</u> is also referred to as **Anabolism**

Stored energy <u>breakdown</u> is also referred to as **Catabolism**

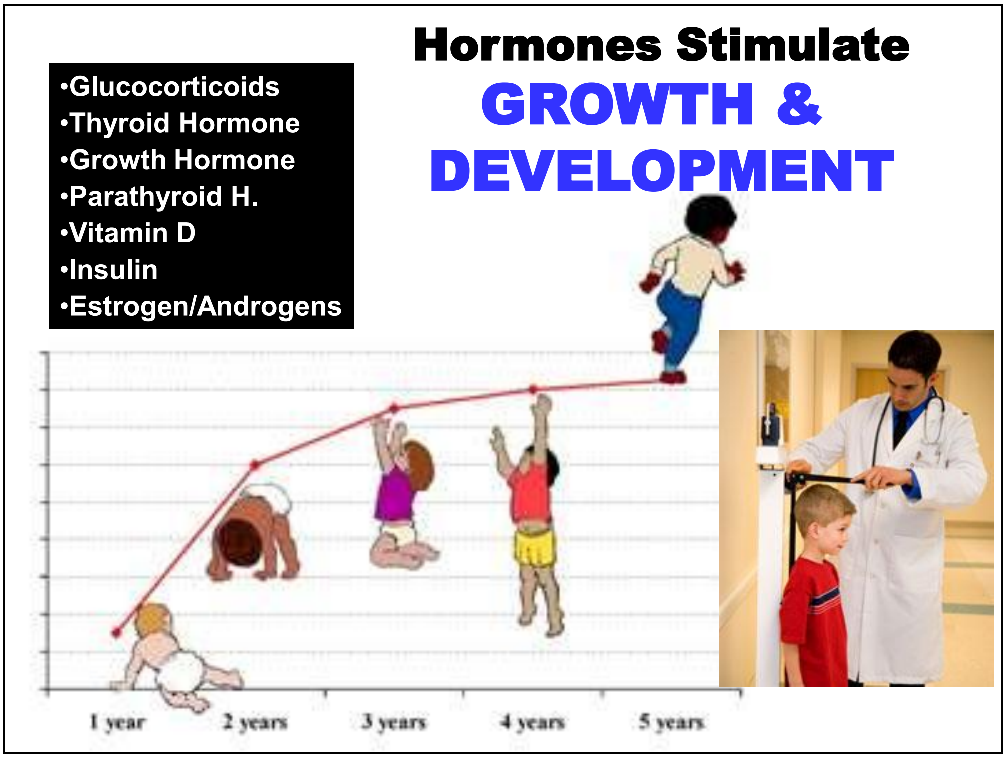
Major Actions of Endocrine Hormones Regulating Metabolic Fuel Processes

	Liver	Muscle	Adipose Tissue
Insulin	+ glycogen synthesis	+ glucose uptake	+ glucose uptake
	+ glycolysis	+ amino acid uptake	+ FFA uptake
	- glycogenolysis	- proteolysis	- lipolysis
	- gluconeogenesis		
	- ketogenesis		
Glucagon	+ glycogenolysis	minimal action	+ lipolysis (minimal)
	+ gluconeogenesis		
	+ ketogenesis		
Cortisol	+ glycogenesis	- amino acid uptake	+ lipolysis
	+ gluconeogenesis	+ proteolysis	- insulin action
		- insulin action	
Growth hormone	+ gluconeogenesis	+ amino acid uptake	+ lipolysis
	+ IGFs/IGFBP	- glucose uptake	- glucose uptake
Epinephrine	+ glycogenolysis	+ glycogenolysis	+ lipolysis
	+ gluconeogenesis	- insulin action	- insulin action
	+ ketogenesis		
Thyroid hormones	+ gluconeogenesis	+ protein catabolism	+ lipolysis
	+ glycogenolysis	+ protein anabolism	+ lipogenesis

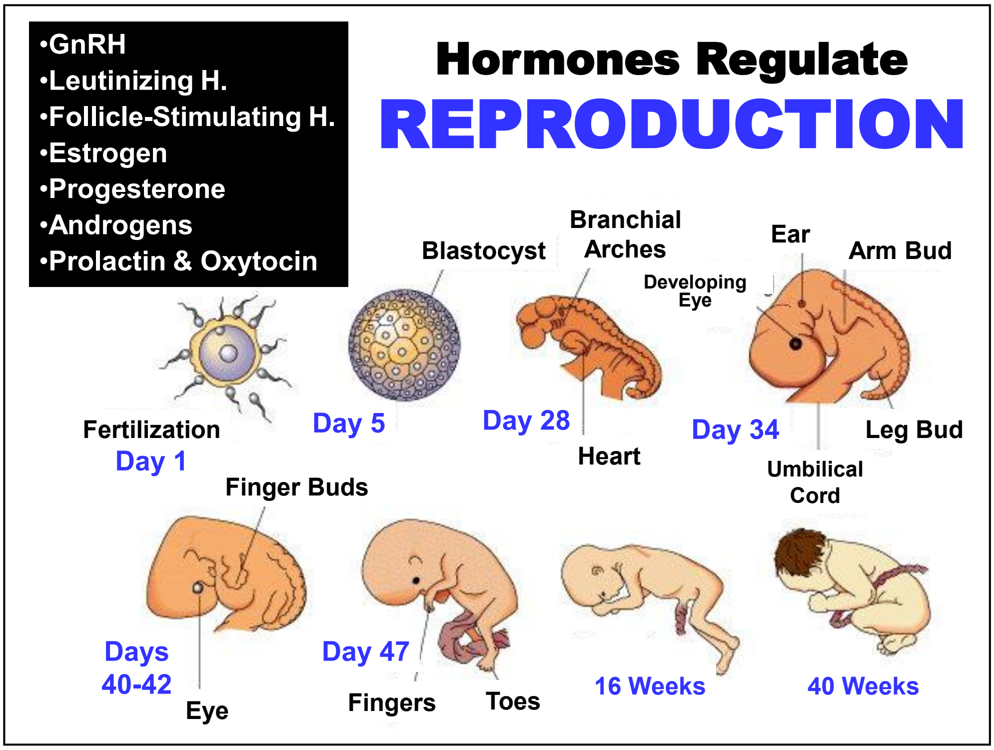
(+) = stimulates; (-) = inhibits

Study Table. Metabolic Actions of Endocrine Hormones.

Summary of diverse and distinct actions of endocrine hormones on metabolic processes, which will be detailed in the relevant section.



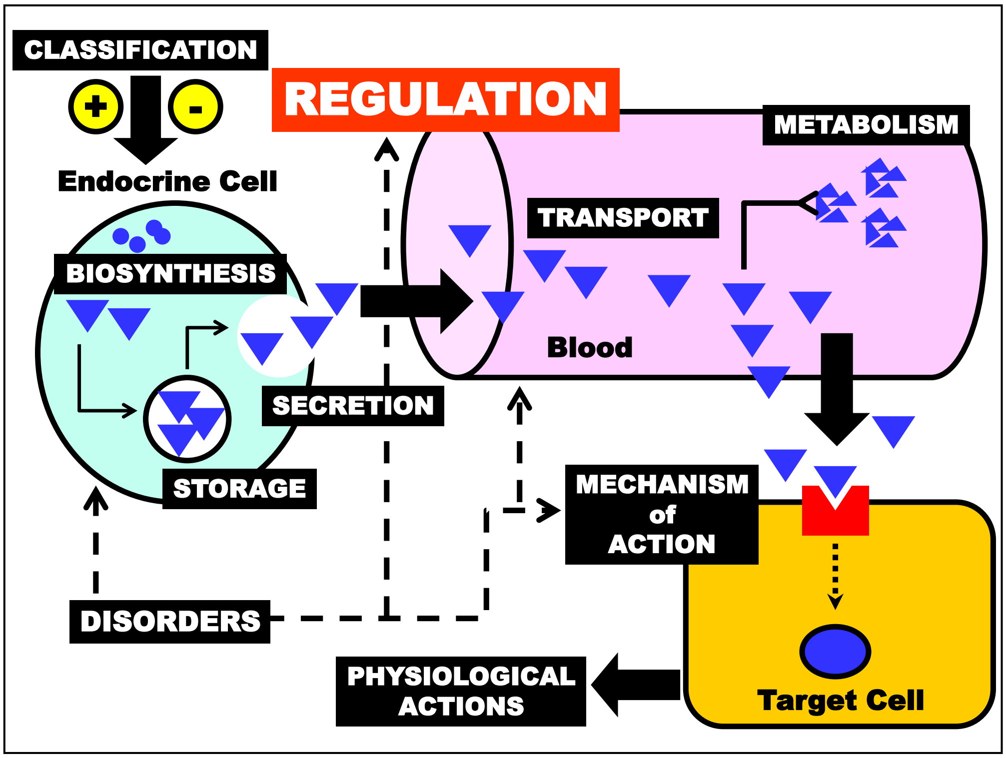
Many hormones also play a vital role in the normal postnatal **Growth & Development** of an individual. Normal physical and mental development are absolutely dependent on endocrine hormones.



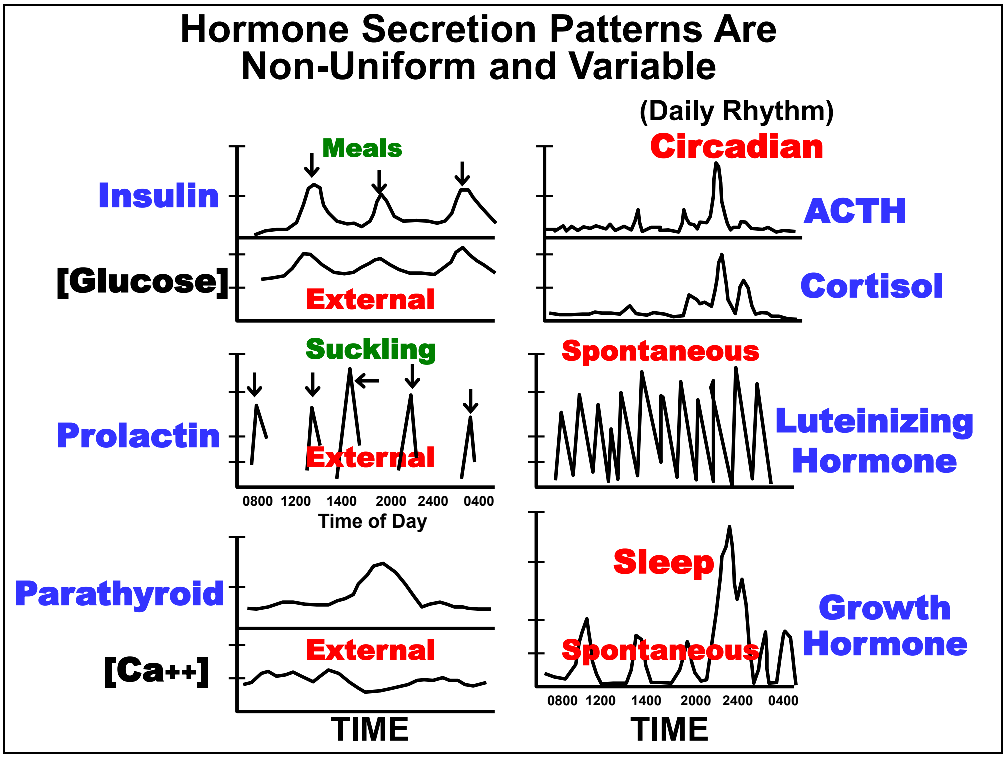
Hormones are essential for all aspects of the Reproduction process from the production of gametes and fertilization to the growth & development of the fetus and ultimately to the attainment of sexual maturity in the adult.

These processes will be detailed during the **Reproduction** section of the course.

However, you will be responsible for knowing the basic endocrine properties of reproductive hormones (e.g. structural class, site of secretion, target tissues, etc.)



Hormone **Secretion Regulation** is discussed next.



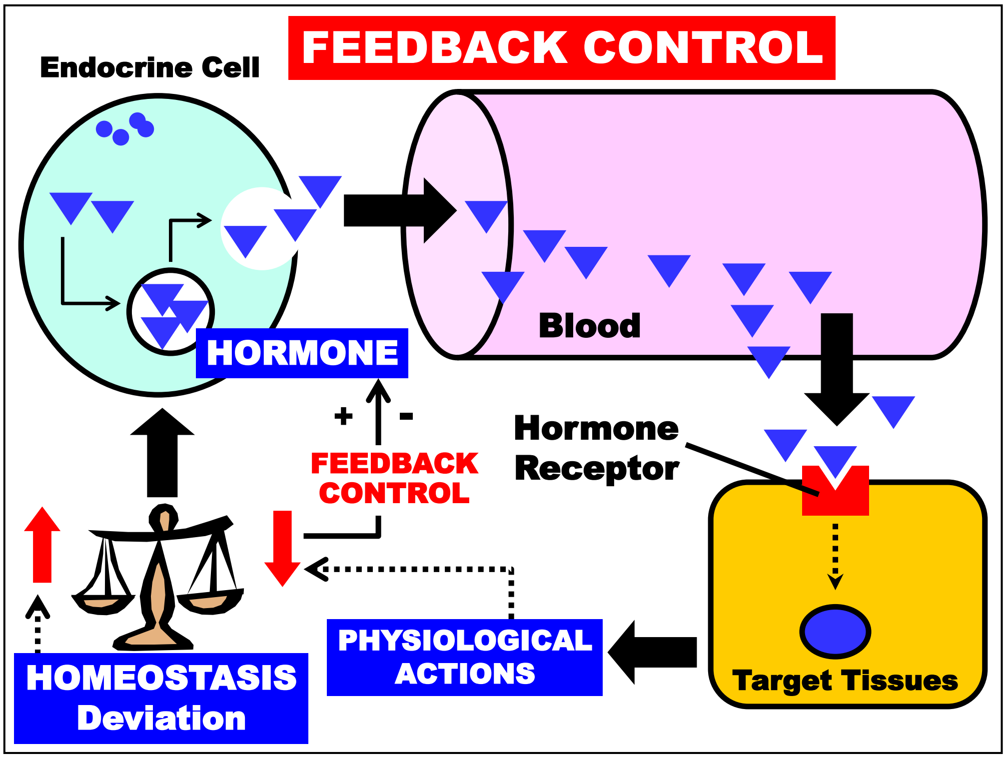
Secretory patterns for different hormones are shown above.

Two concepts to note are that (1) the level of a hormone in plasma (*y-axes*) does **NOT** exist at a **Uniform Concentration** throughout the day (Time *x-axes*), (2) hormone secretion pattern *varies* with each hormone and is determined by specific physiological stimuli.

The clinical importance of these properties is that patient hormone levels can vary significantly depending on (1) <u>when the hormone is measured</u> and (2) the particular <u>physiological condition</u>.

These **variables complicates the ability to assess whether an appropriate level of a hormone is being maintained by a patient.**

Therefore, awareness of this variable and the influencing factors are necessary to appropriately interpret diagnostic data assessing hormone concentration. Furthermore, multiple serial measurements or other more sophisticated techniques may be required to accurately quantify hormone concentration and secretion.



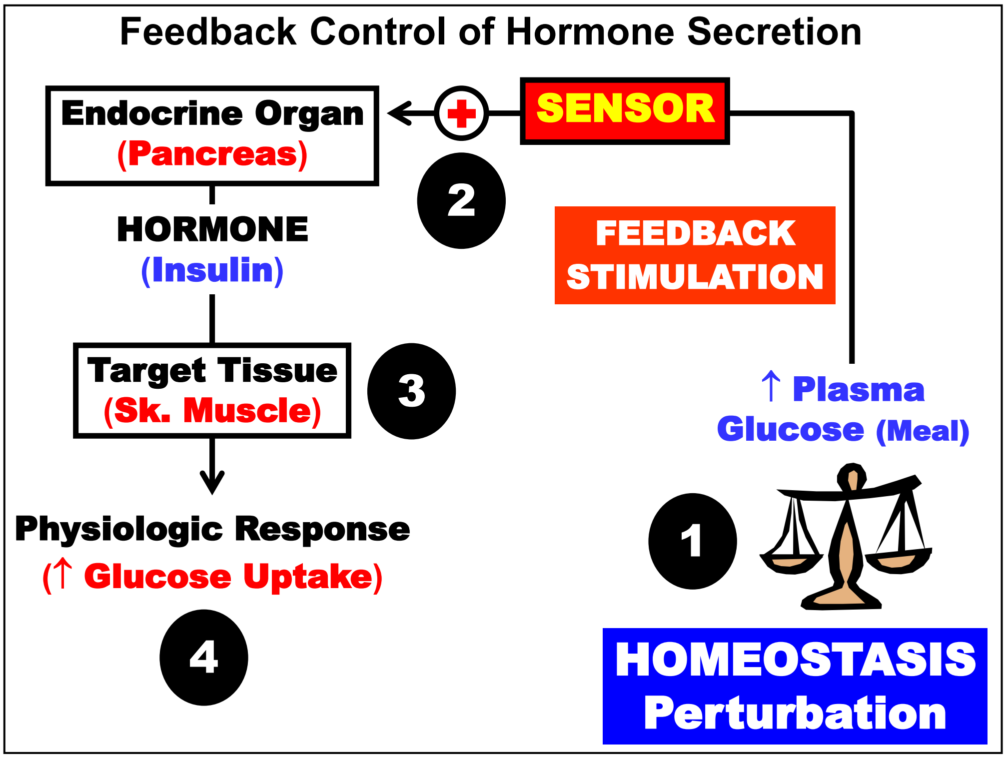
Hormone secretion from Endocrine Cells (*top left*) typically occurs in response to a Homeostatic Deviation (*bottom left*) in a physiological parameter that the hormone regulates.

By definition, endocrine hormones are secreted into the blood, which transports the hormone to various Target Tissues (*bottom right*).

Hormones affect Target Tissues by binding to cellular Hormone Receptors that recognize the specific Hormone.

Target Tissues respond to the Hormone by producing a Physiological Action related to the deviated physiological parameter that acts to regain Homeostasis of the parameter.

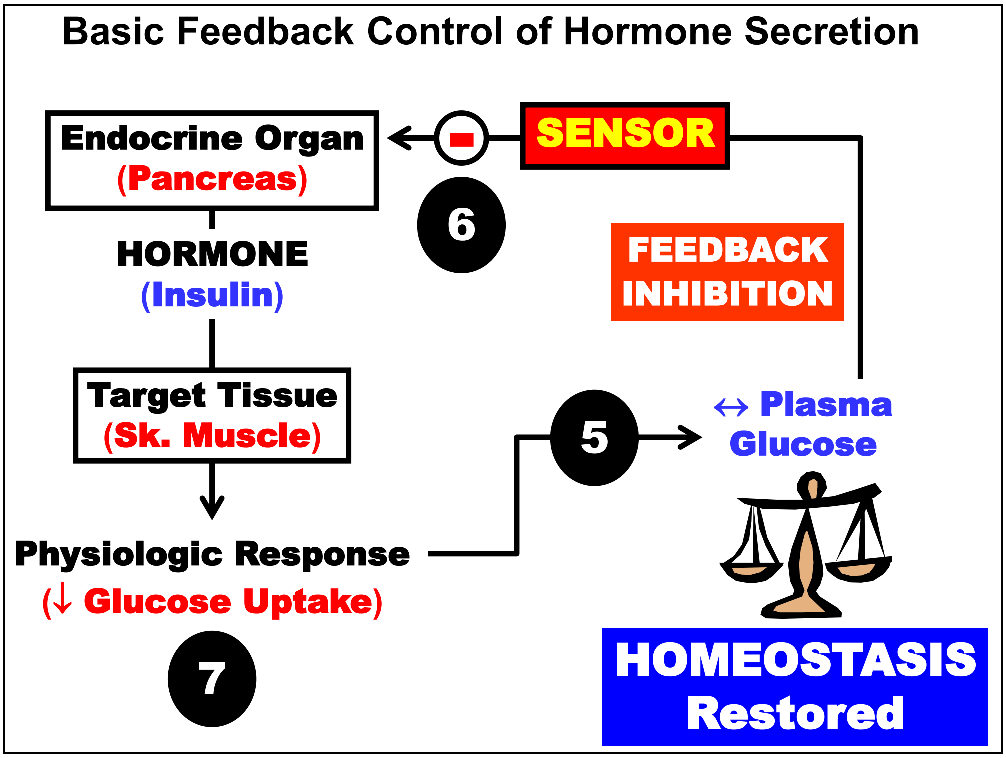
As discussed later, normalization of the parameter usually acts to suppress or shut-off further hormone secretion via a **Feedback Control** (+/-) mechanism.



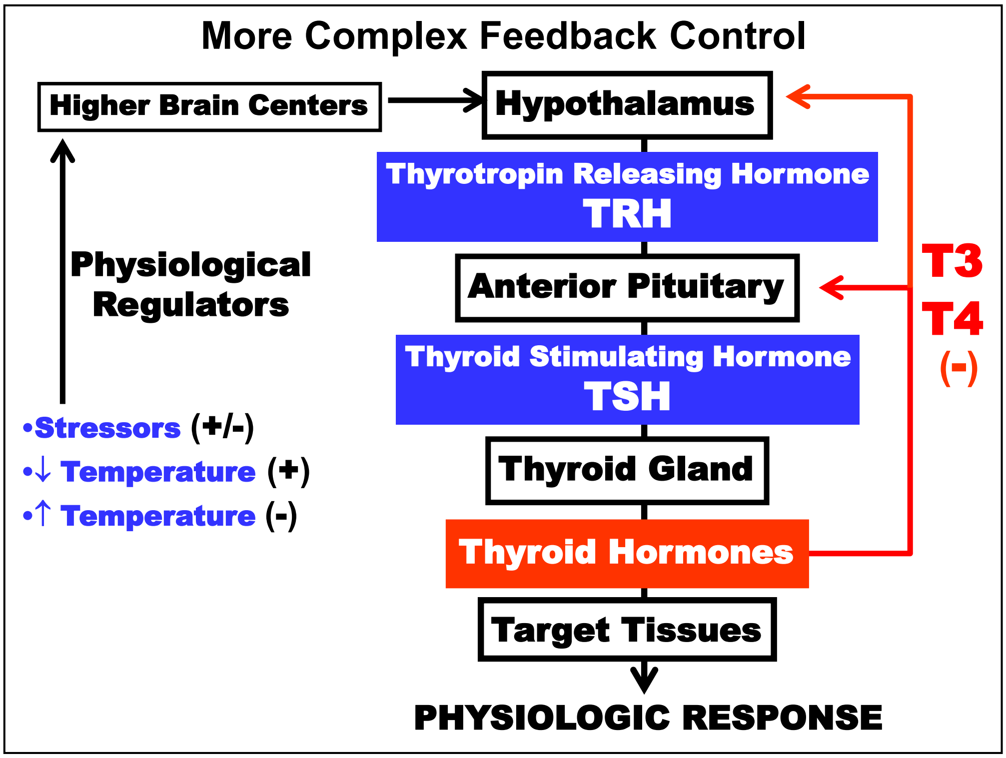
Hormone secretion increases or decreases in response to fluctuations in physiological parameters they regulate.

Feedback Control of Hormone Secretion is an important control mechanism that modulates hormone secretion in response to homeostatic perturbations.

Control of **Plasma Glucose** level by **Insulin** is an example of the simplest control loop as shown above and in the next figure. Regulation is initiated by (**1**) **Homeostasis Perturbation** (*bottom right*) of a physiological parameter (e.g. ↑ Plasma Glucose). (**2**) A **Sensor** mechanism associated with the Endocrine Organ (i.e. Pancreas) detects the perturbation which then stimulates (**+ = Feedback Stimulation**) in this case (or inhibits in other cases) the secretion of the regulatory **Hormone** (Insulin). (**3**) The Hormone next acts on a **Target Tissue** (Sk. Muscle) that (**4**) stimulates a **Physiologic Response** (↑ Glucose Uptake) that normalizes the parameter back to a Homeostatic level.

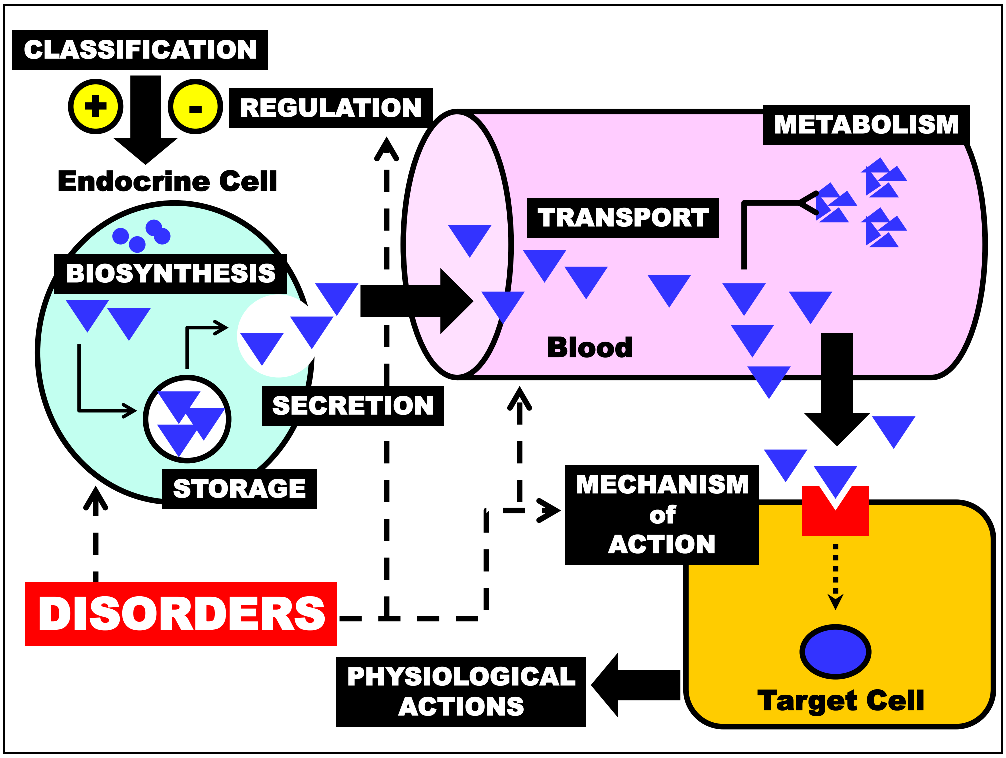


(**5**) If Homeostasis is achieved (↔ Plasma Glucose), (**6**) normalization is detected by the Sensor, which Suppresses (**- = Feedback Inhibition**) further Hormone (Insulin) secretion. (**7**) Stimulation of the Target Tissue (Sk. Muscle) ceases and therefore the Physiological Response (↓ Glucose Uptake) is muted. More complex regulatory loops will be discussed later.



More complex feedback mechanisms regulating endocrine hormone secretion will be discussed later.

The hypothalamic-anterior pituitary control of thyroid hormone secretion is shown above and will be detailed later in the **Thyroid Hormone** section.



Endocrine **Disorders** are discussed next in general terms.

Classic Endocrine Disorders

Pituitary	**Growth H.**	**Deficiency:**	**Dwarfism**
		Excess:	**Gigantism**
			Acromegaly
Thyroid	**Thyroid H.**	**Deficiency:**	**Myxedema**
			Hashimoto's
			Cretinism
		Excess:	**Graves Disease**
Adrenal	**Glucocorticoid**	**Deficiency:**	**Addison's Disease**
		Excess:	**Cushing's Disease**
	Mineralcorticoid	**Excess:**	**Conn's Disease**
	Catecholamine	**Excess:**	**Pheochromocyt.**
Pancreas	**Insulin**	**Deficiency:**	**Type 1 Diabetes**
		Excess:	**Type 2 Diabetes**
Calcium	**Parathyroid H.**	**Deficiency:**	**Hypo-PTH**
		Excess:	**Hyper-PTH**
	Vitamin D	**Deficiency:**	**Osteomalacia**
			Rickets

Endocrine disorders almost always result from **Hyperfunction** (**Excess**) and **Hypofunction** (**Deficiency**) of hormones on target cells.

A number of classic endocrine disorders related to each of the major endocrine organs will be discussed in later sections to emphasize the normal physiological actions of hormones. A summary is listed above of some the major conditions that will be introduced in this section or later during the clinical portion of the course.

Most Endocrine Disorders Generally Related to Either:

EXCESSIVE or DEFICIENT
Secretion or Action

Interpretation of Hormone Level in Blood Can Be Complex…compensation, feedback regulation, receptor defects

…Patient Displays Symptoms of Hormone HYPOsecretion with Elevated Hormone Level e.g. Type 2 Diabetes

Most endocrine disorders involve either **Excessive or Deficient Hormone Secretion or Action** on target cells.

However, interpretation of measured hormone levels in blood can be complex due to pathophysiological factors such a compensatory and regulatory feedback mechanisms as well as hormone receptor impairments.

For instance, a patient can display clinical symptoms consistent with a HYPOsecretion disorder, but the hormone suspected to be deficient may be elevated above in blood.

For example, patients with type 2 diabetes commonly present with **hyperglycemia** (consistent with insulin deficiency) and compensatory **hyperinsulinemia**.

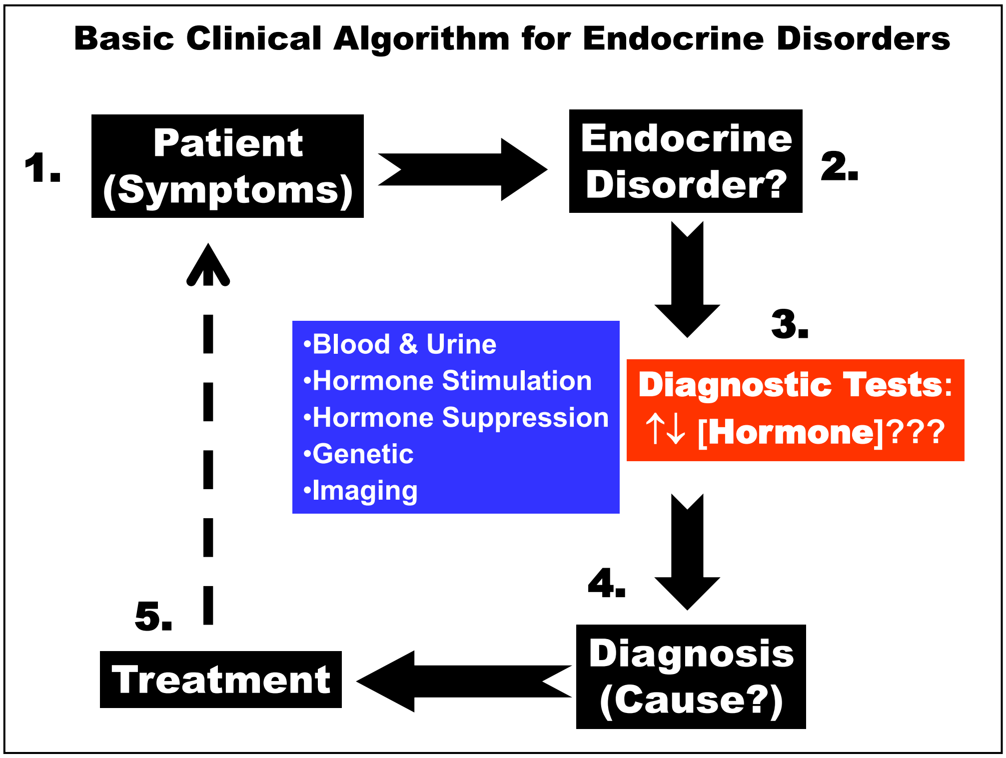


Diagram is the basic clinical algorithm for assessing and diagnosing endocrine disorders.

There are several types of **Diagnostic Tests** available depending upon the suspected etiology.

However, hormone assessment is complicated by natural fluctuations in hormone concentration as shown next.

Narrated PowerPoint Presentations

1. **Overview & Principles of Endocrine System**
2. **Hypothalamic Neuroendocrine Hormones**
3. **Anterior Pituitary Hormones**
4. **Posterior Pituitary Hormones**
5. **Thyroid Hormones Basic Properties**
6. **Thyroid Hormone Disorders**
7. **Parathyroid Hormone**
8. **Vitamin D and Calcitonin**
9. **Adrenal Cortex Hormones**
10. **Steroid Hormone Biosynthesis & CAH Disorders**
11. **Adrenal Medulla Hormones**